Impact of the MW-M31 Merger on the Dark Matter Halo Distribution of M33

Nicolas Mazziotti[1]

[1] *University of Arizona*

1. INTRODUCTION

The idea of dark matter was first hypothesized by F. Zwicky (1933) to explain the enormous velocity dispersion of galaxies in the Coma cluster, far exceeding the dispersion expected from the observed luminous matter. The same conclusion was later reached by V. C. Rubin & W. K. Ford (1970) to account for the higher than expected rotation speeds in the outskirts of M31, introducing the idea that all galaxies are enveloped by a massive halo of dark matter. The inability to observationally detect dark matter other than through gravitational interactions with luminous matter has led to the conclusion that dark matter particles are extremely cold and weakly-interact with each other. This paradigm forms the basis of the Lambda cold dark matter (ΛCDM) theory of cosmology, which asserts that dense regions of dark matter in the early universe provided the necessary scaffolding for galaxy formation by pulling in baryonic matter.

Relating the dark matter halo structure of a galaxy to observable gas and stellar features is essential for understanding the role of dark matter in galaxy evolution and constraining the properties of dark matter particles. Galaxies in the Local Group are ideal candidates to study dark matter halos because of their proximity, especially the halos of likely satellites such as the LMC, SMC, and M33 that would otherwise be too distant to probe. This may lend valuable insight into a longstanding issue in ΛCDM cosmology: the missing satellites problem, which arises from the fact that fewer satellites have been found in the Local Group than predicted (G. Kauffmann et al. 1993).

Efforts to characterize the 3D shape of dark matter halos are largely driven by simulations, from which a common result has been that halos often appear as prolate (elongated along poles), oblate (flattened at poles), or triaxial (see Figure 1). By examining Milky Way sized dark matter halos in the Aquarius cosmological N-body simulations, C. A. Vera-Ciro et al. (2011) found that halo shape measured at the virial radius typically evolved from prolate to a more triaxial/oblate distribution. Narrow filaments of infalling material were found to correlate more with prolate configurations, while isotropic accretion gave rise to triaxial/oblate configurations. However, galaxy interactions may cause significant deviations from these three shapes. In N. Garavito-Camargo et al. (2021), N-body simulations of the LMC passing through the Milky Way showed that asymmetric perturbations were induced in the MW halo that did not fit a prolate, oblate, or triaxial distribution.

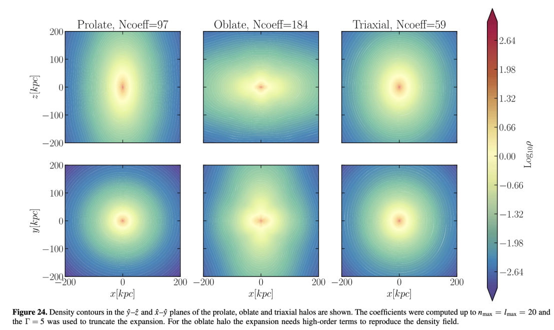

Figure 1. Figure 24 taken from N. Garavito-Camargo et al. (2021) displaying the morphology of prolate (left column), oblate (middle column), and triaxial (right column) dark matter halos. . The top row is the x-z projection while the bottom row is x-y.

The connection between dark matter subhalos and the missing satellites problem is an area of ongoing research. A thorough exploration into the geometrical evolution of satellite sized halos, as opposed to Milky Way sized halos, has yet to be conducted, though simulations of galaxies in the Local Group have grown significantly in resolution. An interesting candidate is M33, as it is the third largest galaxy in the Local Group and lacks a central bulge common for spiral galaxies. Investigating how the shape of M33's dark matter halo evolves as it orbits M31 and the eventual merger between M31 and the MW could probe the stability of such satellite galaxies. A significant shift in halo configuration could be indicative of a process that might have destabilized many satellites in the early universe.

2. PROPOSAL

2.1. *This Proposal*

In this project, I aim to utilize the simulation data provided by G. Besla et al. (2012) of the MW-M31-M33 system to characterize the shape of M33's dark matter halo from the present day until ∼12 Gyr in the future. I would to like to see if M33's halo at the present day can be described by a prolate, oblate, or triaxial distribution, and if this shape changes at all over time, specifically in response to the M31-MW merger.

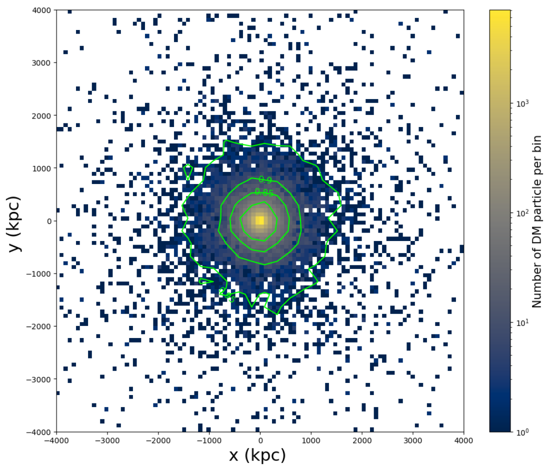

Figure 2. Example density projection (x-y) of the dark matter particles in M31 at Snapshot 0. Contours are overlayed in green. This same plot will be produced in my project for M33.

2.2. *Methods*

Most of my methods are taken from Lab 7. I plan to use the dark matter particle data of M33 in the simulation (particle type 1) to extract information about M33's halo. First, I will use the `CenterOfMass` code to calculate the center of mass of M33 according to the dark matter particles at Snapshot 0 (present day). The dark matter particles are used to calculate the COM as opposed to the disk particles because the halo COM will likely differ from the disk particle COM. I can then calculate positions of all of the dark matter particles in the COM frame. These coordinates will be used to make a projected density plot as in Lab 7 using the `matplotlib` 2D histogram, overlayed with contours using the `density_contour` function (see Fig. 2). This plot should have dense contours like in 1 to see the halo shape more clearly. At this stage, I will attempt to classify the halo shape is either prolate, oblate, or triaxial, if possible. I will generate the same plot for every 10 snapshots up through Snapshot 800, ∼12 Gyr in the future, and then produce an animated movie from the images with Adobe Photoshop to visually check for any changes in the halo shape. If no changes are observed, I will repeat this same process for a different projection, using the `RotateFrame` function from Lab 7 if helpful. I will compare the time evolution of the movies to the M31-M33 and MW-M31 orbital plots in Homework 6 to see if any significant changes occur in response to the MW-M31 merger.

2.3. *Hypothesis*

I hypothesize that the halo distribution of M33 will be more triaxial at the present day because it has not been recently perturbed by M31 or any other object. As the M31-MW merger occurs, I predict that the halo shape of M33 will become more prolate due to the added mass of the Milky Way, elongating M33's halo along the collision axis. I don't think the halo will deviate from a prolate, oblate, or triaxial distribution post-merger because M33 is currently far enough away from M31 and massive enough to not be completely tidally disrupted by the merger. Additionally, dark matter does not interact with itself so the halo shape should remain approximately stable despite the introduction of the MW dark matter halo during the merger.

REFERENCES

Besla, G., Kallivayalil, N., Hernquist, L., et al. 2012, MNRAS, 421, 2109, doi: 10.1111/j.1365-2966.2012.20466.x

Garavito-Camargo, N., Besla, G., Laporte, C. F. P., et al. 2021, ApJ, 919, 109, doi: 10.3847/1538-4357/ac0b44

Kauffmann, G., White, S. D. M., & Guiderdoni, B. 1993, MNRAS, 264, 201, doi: 10.1093/mnras/264.1.201

Rubin, V. C., & Ford, Jr., W. K. 1970, ApJ, 159, 379, doi: 10.1086/150317

Vera-Ciro, C. A., Sales, L. V., Helmi, A., et al. 2011, MNRAS, 416, 1377, doi: 10.1111/j.1365-2966.2011.19134.x

Zwicky, F. 1933, Helvetica Physica Acta, 6, 110